

ADVANCED RED TEAMING – MASTER TECHNICAL EDITION

Offensive Simulation, Detection Engineering & Forensic Analysis

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1. Technical Command Samples (Sanitized Lab Format)

The following command samples are presented in sanitized lab format for academic

documentation only. Cloud Enumeration (Example Format): > aws s3 ls --profile lab-user

IAM Role Inspection (Example Format): > aws iam list-attached-role-policies --role-name

PowerShell Logging Verification: > Get-WinEvent -LogName

"Microsoft-Windows-PowerShell/Operational" Network Connection Review: > netstat -ano |

findstr :443 These samples demonstrate telemetry generation for detection validation, not exploitation.

2. Log Snippet & Detection Evidence Examples

Example: PowerShell Execution Log Entry Event ID: 4104 User: LAB-USER

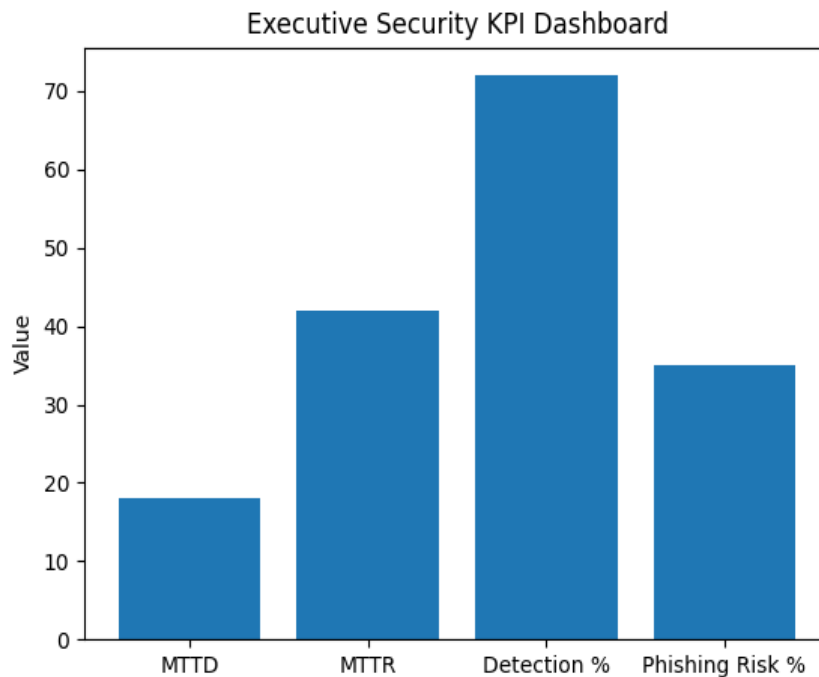
CommandLine: EncodedCommand Detected Severity: High Example: IAM Privilege

Escalation Log EventName: AttachRolePolicy SourceIPAddress: 192.168.1.50 Result:

Success Example: HTTPS Beacon Pattern DestinationPort: 443 Repeated Interval: 60

seconds Anomaly Score: Elevated These snippets illustrate detection engineering validation points.

3. Executive Security Dashboard



KPI Interpretation: • MTTD (18 min) – Moderate detection delay • MTTR (42 min) – Acceptable response time • Detection Coverage (72%) – Requires improvement • Phishing Risk (35%) – High human-factor vulnerability Trend tracking over quarterly simulations is recommended.

4. Risk Scoring Methodology

Risk scores were calculated using a qualitative likelihood $\times$ impact matrix. Likelihood

Factors: • Exploit Simplicity • Exposure Surface • Authentication Controls Impact Factors: •

Data Sensitivity • Privilege Level Obtained • Lateral Movement Potential Scoring Model:

Low (1–2) Medium (3) High (4) Critical (5) IAM privilege escalation scored Critical due to full administrative control exposure.

5. Forensic Artifact Analysis

Endpoint Artifacts Identified: • PowerShell Operational Log Entries • Security Event Log (4624 / 4672) • Scheduled Task Creation Events • Prefetch Metadata Entries • Network Connection Cache Records Cloud Artifacts: • CloudTrail IAM Role Modification Logs • S3 Access Log Entries • Authentication Metadata Memory-Level Indicators (Simulated): • Suspicious Parent-Child Process Chains • Encoded Command Execution These artifacts support digital forensic validation of simulated adversary activity.

6. References & Research Citations

• MITRE ATT&CK; Framework – Enterprise Matrix • NIST SP 800-61 – Computer Security Incident Handling Guide • AWS Security Best Practices Documentation • CISA Joint Cybersecurity Advisories • SANS Incident Response Whitepapers • PTES – Penetration Testing Execution Standard These references guided the adversary emulation and reporting methodology.

7. Final Technical Conclusion

This master-level red team report integrates offensive simulation, defensive analytics, executive KPI visualization, and forensic artifact validation. The assessment demonstrates that detection maturity, identity governance, behavioral analytics, and continuous adversary emulation are critical to achieving cyber resilience. Future improvements should focus on zero-trust architecture, automated detection engineering, and proactive threat intelligence integration.